

Aritra Sadhukhan

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WORK EXPERIENCE

Engineer Web — Codeclouds

Apr 2024 – Present

- Enabled non-engineering teams to launch patient engagement campaigns without code, measured by 7+ reusable automation templates (abandoned-cart retargeting, prescription reminders, admin-configured webhooks), by architecting a headless Python workflow orchestration engine with durable wait states and idempotent, Kafka-driven execution.
- Established a unified audit trail across 5 microservices for a telehealth platform while cutting long-term log storage costs, by building an AWS Firehose/Lambda/Athena pipeline with Parquet compression and automated S3 lifecycle tiering.
- Stabilized a 100,000-product catalog sync that was causing recurring production failures, by re-architecting it onto AWS SQS and Lambda with resilient cursor-based batching that eliminated rate-limit failures and database overload.
- Cut new-service boilerplate by ~40% across 30+ microservices, by designing a standardized TypeScript backend framework for Fastify and Express.js with a CLI code generator producing functional CRUD modules.
- Scaled a high-traffic e-commerce checkout platform to sustain ~30,000 requests/second, by leading Locust-based load testing and resolving bottlenecks through static routing, Redis/in-memory caching, database indexing, and PM2 clustering.
- Improved template rendering throughput by ~5x across horizontally scaled instances, by engineering a two-tier caching architecture with Redis, in-memory LRU caches, Pub/Sub invalidation, and AWS sticky sessions.
- Powered sales, conversion, and fraud-detection dashboards processing ~1M events and 100,000 orders (refreshed with a 1-day data lag), by building a multi-tenant analytics backend on Kafka and MongoDB aggregation pipelines.
- Secured access across internal workspace applications with 4 MFA methods (TOTP, OTP, passkeys, recovery tokens), by building a multi-tenant SSO and identity platform on OAuth 2.0/OIDC and JWT with device-level session management and RBAC.
- Enabled data-driven upsell optimization across ~1M daily checkout requests, by building a configurable A/B and split-testing engine with session-persistent variant assignment.
- Unified checkout and order processing across 5 CRM and e-commerce platforms (Sticky, CheckoutChamp, Vrio, Shopify, WooCommerce), by designing an extensible integration framework with a shared abstraction layer using the Abstract Factory pattern.
- Powered ~1M merchant-customized checkout page requests per day, by designing a custom HTML templating language and synchronous rendering engine that resolves templates live from Amazon S3.

Assistant Engineer Web — Codeclouds

Jun 2022 – Apr 2024

- Simplified HR policy configuration by consolidating 52 leave-accrual combinations into 4 extensible strategies, by architecting a modular accrual engine with cron-driven, idempotent asynchronous processing via Google Cloud Tasks.
- Delivered production REST APIs and microservices across 6 core HR platform modules (leave management, ticketing, notifications, scheduling, knowledge base, centralized logging), by building on Node.js, Express.js, TypeScript, and Kafka.

LANGUAGES & TECHNOLOGIES

Languages	TypeScript (Node.js), Python, Go, SQL, HTML, CSS
Frameworks & Libraries	Express.js, Fastify, FastAPI, TypeORM, Fiber (Go), Socket.IO, NumPy, pandas, scikit-learn, LangChain, LangGraph
Databases	PostgreSQL, MongoDB, DocumentDB, ChromaDB
Cloud & Infrastructure	AWS (S3, Lambda, SQS, Firehose, Athena, ECS, EC2, IAM, ECR), Docker, Kubernetes, Kafka, Redis, Nginx, Linux, GitHub Actions
Architecture & Protocols	Microservices, Event-Driven Architecture, RAG, REST APIs, RBAC, OAuth 2.0 / OIDC, WebRTC, WebSocket

EDUCATION

B.Tech, Computer Science & Engineering — Ideal Institute of Engineering

Jul 2023 – Jul 2026

CGPA: 8.3

Diploma, Electrical Engineering — North Calcutta Polytechnic

Jul 2018 – Aug 2021

CGPA: 9.1

EXTRACURRICULAR

Projects

Hush — hush.swiftgeek.in

End-to-end encrypted chat application with zero-knowledge architecture, AES-GCM/CTR message and media encryption, RSA-based key exchange, and WebRTC audio/video, with a horizontally scalable Socket.IO backend for chunked, encrypted file transfer.

Class-validator-custom-errors — npmjs.com/package/class-validator-custom-errors

NPM package forked from class-validator enabling fully customizable, internationalization-ready error messages for TypeScript applications.

Certifications

TypeScript (2024) · Microservices with Go and gRPC (2025) · Python (2024) · Docker (2024) · MySQL (2025) · AWS Cloud Practitioner (2026)

Open Source

Contributor to [Twenty](#), [OpenBB](#), [Unkey](#), [Dub](#), [Papermark](#), [Openplane](#), and [Hanko](#). Hacktoberfest: Top-12 finisher (2024), [Super Contributor](#) (2025).

Awards

Outstanding Performance Award In Development — Spectrum (2023).